

AJAY CLARK-DESAI

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Engineering (Electrical & Computer Engineering)

Pittsburgh, PA

May 2028

- Dean's List (Fall 2024)

WORK EXPERIENCE

Cisco Systems

Technical Intern

Austin, TX

May 2026 – Aug 2026

- Contributing to the development and reliability of Cisco's VPN platform, focusing on secure connectivity, automation workflows, and infrastructure-level improvements
- Implementing features, debugging system-level issues, and collaborating with engineers across security and networking teams to enhance performance and user experience

Carnegie Mellon University

Network Operations Student Technician

Pittsburgh, PA

Sep 2025 – May 2026

- Deployed and maintained campus-wide networking infrastructure (switches, routers, structured cabling) supporting 50+ academic and administrative buildings
- Monitored network performance and resolved 100+ service tickets, ensuring consistent uptime for thousands of users
- Applied enterprise networking protocols and real-time diagnostics to support IT operations in a high-demand environment

Rice University

Research Assistant, Developer

Houston, TX

May 2025 - Aug 2025

- Optimized signal processing workflows in C#, enabling precise software control for quantum squeezing experiments.
- Enhanced CUDA kernels and profiled with Nsight Compute, achieving a 50% performance improvement in GPU operations
- Contributed to research on quantum optics and error correction in photonic systems, advancing experimental computing methods

RELEVANT COURSEWORK

Analog Circuits • Microelectronic Circuits • Digital Logic • Data Structures & Algorithms • Semiconductor Fundamentals • Quantum Physics • Discrete Math

PROJECT EXPERIENCE

Guitar Effect Pedals (*Build 18, CMU*)

Dec 2025 – Jan 2026

- Engineered supplier-ready PCB designs replicating guitar effect circuits (overdrive, EQ), with complete BOM/CPL documentation and assembly process specifications to meet professional fabrication standards

Brain Scanning Video Game Controller (*Senior Capstone*)

Sept 2023 - May 2024

- Developed Python software to interpret EEG signals from Muse Brain Scanner, improving accuracy with signal processing
- Integrated software into a game to provide individuals without fine motor control an ability to play video games by blinking

ACTIVITIES & LEADERSHIP

CIA Buggy (*Club*)

Assistant Build Lead

Pittsburgh, PA

Aug 2024 - Present

- Assistant Build Lead for competitive racing team designing and fabricating resin-infused carbon fiber buggies
- Coordinated build sessions and mentored new members in fabrication techniques
- Directed development of GPS-based animation software to visualize buggy movement for race analysis and performance optimization.

SKILLS

Programming: Java, Python, C#, C, C++, Rust, MATLAB, HTML/CSS/JavaScript, CUDA, System Verilog

Hardware & EE: PCB Design & Fabrication, FPGA Design, Microelectronic Circuit Design, Signal Processing

Tools: GitHub, Cadence, Vivado, Linux, Autodesk Fusion, Microsoft Office

Language: English (Native), Chinese (Elementary)